

ECE 7410: Image Processing

Lab 1: Geometric Transformation and Histogram Equalization

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Geometric Transformation

1. Download the test image (img1.png) from Brightspace under Lab 01. Read the image, convert the image to a grayscale image, and display the image.

```
imc = imread("./img1.png");  
  
img = rgb2gray(imc);  
  
imshow(img);  
title('Image in grayscale');
```

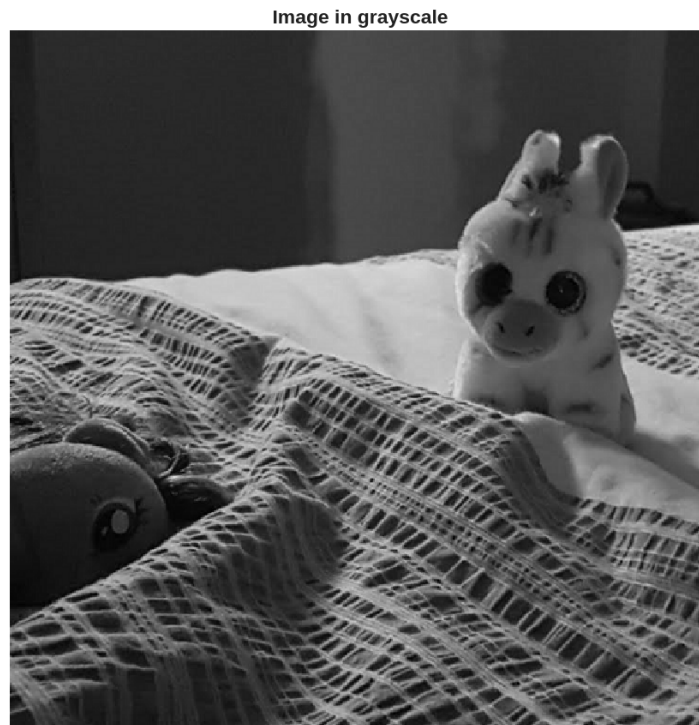


Figure 1: img1.png in greyscale

2. a. Define the transformation matrix with a rotation angle of 45 degrees

```
theta = 0.25*pi;  
R = [cos(theta) sin(theta) 0; ...  
     -sin(theta) cos(theta) 0; ...  
     0 0 1];
```

- b. Calculate the size of the transformed image and generate an empty image with the calculated size

```
[y_max, x_max] = size(img); % Obtaining the size of the image  
corners = [0,0,1; ...  
           x_max,0,1; ...  
           0,y_max,1; ...  
           x_max,y_max,1]; % Corner points of the image  
  
new_corners = corners*R % New location of corners  
new_width = round(max(new_corners(:,1)) - ...  
                  min(new_corners(:,1)));  
new_height = round(max(new_corners(:,2)) - ...  
                   min(new_corners(:,2)));  
new_img = zeros(new_height,new_width);
```